

Miles Franck

📍 Silver Spring, MD

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I am a software engineering graduate with a technical foundation in Python, database systems, and secure software development. Hands-on experience across SOC operations, automation tooling, and network systems — with exposure to SIEM tools, vulnerability analysis, and threat modeling. CompTIA Security+ in progress. Looking to grow within a collaborative engineering team and apply a security-aware mindset wherever it adds value.

EDUCATION and SCHOLARSHIPS

- B.S Software Engineering | Allegheny College 2021-2025
- College Trustee Scholarship Recipient 2021-2025
- Additional Merit Scholarship Recipient 2022-2024

EXPERIENCE

Carnegie Mellon Heinz College Summer

Fellowship—Carnegie Mellon University *Summer Fellow* / 2025

- Completed a hands-on IT and cybersecurity fellowship centered on real-world technical challenges and security workflows
- Applied cybersecurity principles and technical problem-solving skills in team-based exercises and projects
- Collaborated with industry professionals on system architecture, security solutions, and project management initiatives

Technical Leader — Allegheny College 2022-2025

- Led technical guidance and support for students and faculty, translating academic and operational needs into effective technical solutions.
- Held office hours and provided one-on-one support to students, assisting with debugging, technical concepts, and project implementation.
- Collaborated with faculty and communicated across both technical and non-technical stakeholders to ensure grading workflows and project outcomes aligned with expectations.

CERTIFICATION

- **CompTIA Security+** — CompTIA (*Expected August 2026*)
- **SOC Operations Analyst II** — BCR Cyber (*Dec 2025*)
- **SOC Operations Analyst I** BCR Cyber (*Nov 2025*)

CORE COMPETENCIES

- Secure SDLC
- Threat modeling
- Network architecture
- Security analysis
- Cryptography fundamentals
- REST API integration
- Data sanitization
- Agile / Scrum collaboration
- Vulnerability assessment
- Technical documentation

SOFTWARE SKILLS

Proficient: Python • SQL • PostgreSQL • MySQL • Git • WSL • Shodan

Familiar with: Nmap • Splunk • Wireshark • Rust

PROJECTS

Rust Server (In Progress) 2025

- Building a multi-client TCP chat server in Rust using the Tokio asynchronous runtime, supporting concurrent client connections and real-time message broadcasting.
- Implemented a publish-subscribe messaging system using Tokio broadcast channels to efficiently distribute messages to all connected clients. salts
- Designed a room management system with unique join codes, user authentication, connection limits, and participant tracking.

BanjoDB 2025

- Built a dual-language architecture separating a secure Rust core from a Python interface to emulate enterprise-style backend isolation
- Mitigated Credential theft risks by implementing Argon2 Hashing with unique per-user salts

- Designed a secure data-at-rest architecture by storing binary file data inside a private SQLite backend with application-level authentication to restrict unauthorized access

Deepseeking-Vulns 2025

- Developed an AI-assisted vulnerability assessment workflow using Shodan, Ollama, and DeepSeek to analyze exposed network services and support penetration-testing preparation and risk analysis

Automated Vulnerability Analysis Tool 2024

- Built a Python tool that queries the Shodan API to enumerate internet-facing assets and cross-references results against CVE/CPE vulnerability databases.